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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPES200- FUNDAMENTALS OF KINESIOLOGY AND BIOMECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes

Maximum Marks: 50

Part A

MCQ

Answer all questions

Each question carries 1 marks

1.How does kinesiology contribute to injury prevention in sports ? [A] / [CO2]

Ans : a) By identifying risk factors and improving movement patterns

2.What is the primary action of the rectus abdominis muscle? [K] / [CO3]

Ans: b) Flexion of the trunk

3.What is the benefit of biomechanical analysis for coaches? [K] / [CO4]

Ans: a) Improved understanding of athlete movement patterns

4. What is one of the applications of kinesiology? [U] / [CO2]

Ans: d) All of the above

5. Which of the following is a benefit of biomechanical analysis in physical education? [K] / [CO4]

Ans : d) All of the above

6. What is the insertion of the biceps brachii muscle? [K] / [CO3]

Ans: b) Radial tuberosity and bicipital aponeurosis

7. What is the origin of the gluteus medius muscle? [K] / [CO3]

Ans: a) Ilium

8. Which of the following is a unit of measurement for velocity? [U] / [CO5]

Ans: a) Meters per second (m/s)

9. What is the unit of measurement for acceleration? [A] / [CO6]

Ans: b) Meters per second squared (m/s^2)

10. How does kinesiology help improve athletic performance? [U] / [CO2]

Ans: a) By optimizing training programs and techniques

[1×10 = 10]

Part B

Short Answer Questions

Answer any 10 questions

Each question carries 2 marks

11. Provide an example of flexion and extension in the human body. [K] / [CO1,CO2]

Ans :

- Flexion: Bending the elbow.
- Extension: Straightening the elbow.

12. What is the primary role of biomechanics in sports? [U] / [CO4]

Ans:

- To analyze movement efficiency
- improve performance
- prevent injuries.

13. Describe the origin of the trapezius muscle. [K] / [CO3]

Ans:

Origin: Occipital bone, spinous processes of cervical and thoracic vertebrae

14. Describe the origin of the adductor longus muscle. [U] / [CO3]

Ans:

Origin: Body of the pubis (inferior to the pubic crest).

15. What is torque in biomechanics? [K] / [CO5]

Ans :

Torque is the measure of the rotational force that cause an object to rotate or turn around a pivot point or axis.

Torque= Force × Movement arm

Torque unit measurement - Newton- meters (Nm)

16. What is abduction in the context of human movement? [K] / [CO1,CO2]

Ans:

Movement of a limb away from the midline of the body (e.g., lifting arm sideways).

17. State the main functions of the hamstrings. [K] / [CO4]

Ans:

• Flexion of the knee and extension of the hip.

• Helps in Walking, Running, Jumping

18. Why is understanding force and momentum important in sports? [U] / [CO6]

Ans:

- It helps improve performance
- prevent injury by optimizing movement
- Helps coaches to analyse mechanical movement

19. Name two muscles of the rotator cuff. [K] / [CO3]

Ans:

Supraspinatus, Infraspinatus (others: Teres minor, Subscapularis).

20. What is the unit of measurement for velocity? [U] / [CO5]

Ans:

Meters per second (m/s).

21. What is circumduction in the context of human movement? [K] / [CO1,CO2]

Ans:

A circular movement that combines flexion, extension, abduction, and adduction (e.g., arm circles).

22. Mention the origin and insertion of Pectoralis Major. [K] / [CO3,CO4]

Ans:

Origin: Clavicle, sternum, and costal cartilages of upper ribs.

Insertion: Bicipital groove of the humerus

[2×10 = 20]

Part C

Short Essay Question

Answer any 4 questions

Each question carries 5 marks

23. Discuss the laws of motion and their application. [A] / [CO5]

Ans:

Newton's First Law (Law of Inertia):

A body remains at rest or in uniform motion unless acted upon by an external force.

Application: A football remains still until kicked.

Newton's Second Law (Law of Acceleration):

Force = Mass × Acceleration (F = ma).

Application: The harder a player kicks a ball, the faster it moves.

Newton's Third Law (Action–Reaction):

For every action, there is an equal and opposite reaction.

Application: When a swimmer pushes water backward with their arms and legs, the water pushes the swimmer forward

24. Explain the historical development of kinesiology from ancient times to the modern era. [U] / [CO1]

Ans:

Kinesiology is the study of movement behaviour of all living organisms. The word kinesiology derived from two Greek words. Kinein - to move, logos - To discourse or study. The study of kinesiology began in ancient Greece, where scholars like Aristotle and Hippocrates studied human movement and anatomy to understand health and performance. During the Renaissance, scientists such as Leonardo da Vinci and Galileo Galilei analyzed body mechanics using observation and measurement. In the 19th century, advances in anatomy and physiology gave kinesiology a scientific base. In the 20th century, with the development of biomechanics, exercise physiology, and motor learning, kinesiology evolved into a distinct scientific discipline. Today, it integrates modern tools such as motion analysis, electromyography, and computer modeling to improve health, fitness, and athletic performance.

25. List the muscles forming the erector spinae group and describe their anatomical arrangement. [K] / [CO3]

Ans:

The erector spinae is a large muscle group located on each side of the vertebral column that helps maintain posture and extend the spine. It consists of three major columns of muscles arranged from

medial to lateral:

1. Spinalis – the most medial group, closest to the spine.

2. Longissimus – the intermediate group, running along the vertebral column.

3. Iliocostalis – the most lateral group, attaching to the ribs.

These muscles extend from the sacrum and iliac crest up to the skull. Together, they help in extension, lateral flexion, and rotation of the vertebral column and are essential for maintaining an upright posture.

26. Discuss the importance of supination and pronation movements in daily activities. [U] / [CO2]

Ans:

Supination is the outward rotation of the forearm or foot, while pronation is the inward rotation. In the forearm, these movements allow the palm to face up (supination) or down (pronation). They are essential in everyday actions such as turning a doorknob, using a screwdriver, typing, eating, or carrying objects. In the lower limb, controlled pronation and supination help with balance, shock absorption, and efficient walking or running. Proper coordination of these movements prevents injuries and supports smooth motor function in daily life and sports.

27. Discuss the importance of the hip flexors and gastrocnemius muscles in daily activities. [U] / [CO3]

Ans:

The hip flexors (including the iliopsoas, rectus femoris, and sartorius) are responsible for lifting the thigh toward the body. They play a key role in walking, running, climbing stairs, and sitting. Weak hip flexors can cause poor posture and movement inefficiency.

The gastrocnemius, located in the calf, assists in plantar flexion of the foot (pointing the toes) and flexion of the knee. It is vital for pushing off the ground while walking, running, or jumping. Together, these muscles ensure mobility, stability, and balance during everyday and athletic activities.

28. Describe the different types of joints in the human body. [K] / [CO5]

Ans:

Joints are the points where two or more bones meet and allow movement. They are classified into three main types:

1. Immovable joints (Fibrous joints): No movement; bones are connected by fibrous tissue. Example: sutures in the skull.

2. Slightly movable joints (Cartilaginous joints): Limited movement; bones are joined by cartilage. Example: intervertebral discs in the spine.

3. Freely movable joints (Synovial joints): Allow a wide range of movement and contain synovial fluid. Examples.

Ball and socket joint (shoulder, hip)

Hinge joint (elbow, knee)

Pivot joint (neck)

Gliding joint (wrist, ankle)

Saddle joint (thumb)

Condyloid joint (wrist joint).

These joints together enable the body’s flexibility, motion, and coordination.

[5×4 = 20]